MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION April 2026

second SEMESTER

Minor Pathway Courses (MPC) - MG2MPCTFD100 - TEXTILES SCIENCE

(2025 ADMISSION ONWARDS)

ANSWER KEY

Part A – Short Answer Questions

1. Why is jute considered eco-friendly?

Jute is considered eco-friendly because it is biodegradable and natural. It is widely used as an eco-friendly fibre.

2. What are fibre identification tests?

Fibre identification tests are tests used to identify different textile fibres based on their physical and chemical properties.

3. Why is silk called the "Queen of Fibres"?

Silk is called the "Queen of Fibres" because of its excellent lustre, strength, elasticity, smoothness and expensive nature.

4. Why is flax suitable for summer clothing?

Flax is suitable for summer clothing because it has excellent absorbency, good air permeability and conducts heat well, giving a cool feeling in hot climates.

5. Define monofilament yarn.

Monofilament yarn is a single strand of filament yarn and cannot be separated as it is an indivisible component.

6. List any two limitations of handloom.

1. Low production speed
2. Requires skilled labour

7. What does 1X3 weft rib indicate?

1X3 weft rib indicates that the weft passes over or under three consecutive warp yarns at regular intervals.

8. Define purl knitted fabric.

Purl knitted fabric is a knitted fabric in which loops are formed on both sides, producing a reversible fabric with high elasticity.

9. What is sizing?

Sizing is the process in which size paste is applied to warp yarns to increase strength and reduce abrasion during weaving.

10. Why are vat dyes insoluble in water initially?

Vat dyes are initially insoluble in water because they must first be converted into a soluble form using a strong reducing agent before dyeing.

11. What are the general fabric requirements for proper printing?

- Fabric should have proper face on top and uniform width.
- Desizing is necessary.
- Singeing removes hairy fibres.
- Scouring improves dye absorption.
- Fabric should be thoroughly dried and free from creases.

12. What is misregistration (off-register printing)?

Misregistration or off-register printing is the improper alignment of different coloured prints in a design during printing.

Part B – Short Essay Questions

13. Describe the origin, composition and main producers of cotton fibre.

Cotton is a natural seed type vegetable fibre called the “King of Fibres.” It is composed mainly of cellulose (90–94%). Cotton fibres are obtained from the seeds of the cotton plant. Main producers of cotton are USA, Russia, Brazil, Egypt, India, Pakistan and China. Egypt produces the best quality cotton.

14. Describe the physical properties of wool fibre such as colour, strength, elasticity and crimp.

- **Colour:** Cream, yellow and grey-yellow.
- **Strength:** Wool has poor strength with tenacity of 1–2 gpd. Strength decreases by 25% on wetting.
- **Elasticity:** Wool has excellent elasticity due to its wavy structure and scales. It can stretch up to 30% before breaking.
- **Crimp:** Wool has natural crimp or waves which increase cohesiveness, resiliency and insulation.

15. Explain seed, leaf and bast fibres with examples.

- **Seed fibres:** Fibres collected from seeds or seed cases. Example: cotton and kapok.
- **Leaf fibres:** Fibres collected from leaves. Example: sisal and agaves.
- **Bast fibres:** Fibres collected from the skin or bast surrounding the stem of plants. They have high tensile strength. Examples: flax, jute, hemp and ramie.

16. Explain cotton count (Ne) and metric count (Nm).

Cotton Count (Ne)

Cotton count is an indirect numbering system. It is defined as the number of hanks weighing one pound of yarn.
1 hank = 840 yards.
As the count increases, the yarn becomes finer.

Metric Count (Nm)

Metric count is also an indirect numbering system. It is defined as the number of 1000 metre units weighing one kilogram of yarn.
As the count increases, the yarn becomes finer.

17. Explain the construction of novelty yarn.

Novelty yarns are yarns with intentional irregularities in size, twist and colour effects.

Construction includes:

- **Base yarn:** Provides strength and stability.
- **Effect yarn:** Gives texture and decorative appearance.
- **Binder yarn:** Holds the effect yarn to the base yarn.

Novelty yarns enhance texture, styling and surface appearance of fabrics.

18. Explain why technical specifications are used instead of common names in the woven fabric industry.

Common names of fabrics may vary in meaning from one country to another, causing confusion in trade. Therefore, technical specifications are used for accurate identification and communication.

Technical specifications include:

- Fabric width
- Ends per inch and picks per inch
- Weave type
- Fibre content
- Warp and weft count
- GSM
- Type of finish applied

These provide clear and uniform information about the fabric.

19. Explain warp knitting and state its advantages and limitations.

Warp knitting is a knitting process in which each needle forms loops with a separate warp yarn running lengthwise in the fabric.

Advantages

- High production speed
- Fabrics are dimensionally stable
- Less shrinkage
- Strong and durable fabrics

Limitations

- Expensive machinery

- Complex process
- Less elasticity compared to weft knitting

20. Describe the classification of looms based on picking mechanism.

Based on picking mechanism, looms are classified as:

1. **Shuttle looms** – picking done by wooden shuttle.
2. **Projectile looms** – picking done by projectile device.
3. **Rapier looms** – picking done using rapier rods.
4. **Air-jet looms** – picking done using compressed air.
5. **Water-jet looms** – picking done using water jets.

21. Discuss screen printing in detail including process, types, characteristics, advantages and disadvantages.

Screen printing is the most widely used method of textile printing. Screens made of silk or nylon mesh stretched over wooden frames are used.

Process

- Screen is prepared by blocking non-design areas.
- Screen is placed over fabric.
- Printing paste is spread using a squeegee.
- Dye passes through open design areas onto fabric.

Types

- Manual flat screen printing
- Mechanical flat screen printing
- Rotary screen printing

Characteristics

- Suitable for multicolour designs
- Faster than block printing
- Good print quality

Advantages

- Can print large designs
- Suitable for many fabrics
- Better colour application

Disadvantages

- Slower than roller printing
- Separate screen needed for each colour

- More costly for small designs

Part C – Essay Questions

22. Explain absorbency, lustre, flexibility and abrasion resistance of textile fibres with suitable examples.

Absorbency

Absorbency is the ability of a fibre to take up moisture. It is expressed as moisture regain percentage. Fibres with good absorbency are comfortable and dye well. Cotton has excellent absorbency due to high cellulose content and air-filled lumen. Wool can absorb up to 20% moisture without feeling damp.

Lustre

Lustre is the shine or brightness of a fibre caused by reflection of light. Smooth and round fibres reflect more light. Silk has excellent lustre because of its smooth surface and triangular cross-section. Cotton has little or no lustre.

Flexibility

Flexibility or pliability is the ease with which fibres bend or twist. Flexible fibres can be easily spun into yarns and resist breaking during folding. Wool and silk are flexible fibres, while flax is comparatively stiff.

Abrasion Resistance

Abrasion resistance is the ability of a fibre to withstand rubbing during use. Toughness, smoothness and pliability improve abrasion resistance. Flax has good abrasion resistance. Fibres with high abrasion resistance produce durable fabrics.

23. Explain the complete process of yarn manufacturing from fibre to yarn.

The manufacturing of spun yarn involves several stages:

1. Blow Room Process

Fibres from bales are opened, cleaned, blended and impurities removed.

2. Carding

Fibres are individualized and cleaned using carding action. A soft strand called sliver is formed.

3. Drawing

Several slivers are combined and drawn to improve fibre alignment and uniformity.

4. Combing

Short fibres, neps and impurities are removed. Fibres become more parallel and uniform.

5. Roving

Sliver is attenuated and given slight twist to form roving.

6. Spinning

Roving is drafted further and twist is inserted to produce yarn. The yarn is wound onto packages.

24. Explain the importance of synchronization of shedding, picking, and beat-up in fabric production.

Fabric production on a loom depends on proper synchronization of the three basic loom motions:

Shedding

Warp yarns are separated to form a shed through which weft yarn passes.

Picking

Weft yarn is inserted through the open shed.

Beat-up

The reed pushes the newly inserted weft into the fabric structure.

Importance of Synchronization

- Ensures smooth weaving process
- Prevents yarn breakage and fabric defects
- Maintains correct fabric structure and density
- Improves production efficiency and fabric quality
- Proper timing ensures accurate interlacement of warp and weft yarns.

25. Explain block printing and discharge printing in detail.

Block Printing

Block printing is a direct style and hand printing process using engraved wooden blocks.

Process

- Design is carved on wooden blocks.
- Fabric is prepared and dye spread on tray.
- Block is dipped into dye and pressed onto fabric repeatedly.
- Fabric is dried and fixed.

Advantages

- Decorative and artistic prints
- Eco-friendly when natural dyes are used
- Unique handcrafted appearance

Disadvantages

- Slow and labour-intensive
- Expensive due to hand process
- Difficult to obtain shading effects
- Gaps may remain between prints

Discharge Printing

Discharge printing is a style in which colour is removed from selected areas of a dyed fabric using chemicals.

Process

- Fabric is first dyed completely.
- Discharge chemical is printed on design areas.
- Chemical destroys the colour in selected areas.
- White or light-coloured designs appear on dark background.

Characteristics

- Sharp outlines
- Attractive contrast designs
- Used for dark background fabrics